The time is ripe to mine Bitcoins! While the statement would've been more accurate had it been said two years ago, it still is a good time to start mining. Now if you don't understand what mining is then the most simple way to put it would be that it is one of the ways of obtaining Bitcoins and its variants i.e. Litecoins and Dogecoins. Anyone with a computer can mine Bitcoins, the process is that simple. You only need a client and you have to let it run continously for a while till you start generating coins. Or so we may think, but little did you know that with each waking moment the Bitcoin mining process keeps on becoming incrementally more difficult. So the more time you take to get in on the action the lesser of a profit you'll be able to make.

## Bitcoin generation

Basically, you are crunching an awful lot of numbers to get Bitcoins as a reward in return. But since it is digital you'll need some mechanism so that people just don't copy a set of bitcoins and then try to spend it all over again. Afterall, double spending is a characteristic of digital currency. This is where the whole proof-of-work function comes into the picture. Bitcoin uses the hashcash proof-of-work function as the backbone of its mining algorithm. Every mining software that is working endlessly is trying to create a proof-of-work which makes their work unique and thus entitles them to a set of coins. Bitcoin uses an algorithm called hashcash to generate proof-of-work. It is costly in terms of time and energy consumed to produce for the pre-defined parameters. But at the same time it shouldn't require as much time to verify a proof-of-work as it takes to produce it. This is why the entire process revolves around scanning for a value that when it is hashed two times with the SHA-256 hashing algorithm, the resulting hash begins with a pre-defined number of zero bits. The average effort needed to hash such numbers increases exponentially with an increase in the number of zero bits required. This same generated hash however, can be easily verified by rehashing the number with a single pass of the double SHA-256 algorithm. So it is easy to verify but difficult to generate.

Then a simple question arises, if it was difficult to create for one machine then maybe two can do the job better. This is undoubtedly true, adding more hardware to the mix will give you more bitcoins but only for a short while. When you increase the number of zero bits you increase the difficulty. And the Bitcoin network has its difficulty parameter readjusted every two weeks so that as time passes and more powerful computers join the